

llama3_8b (meta-llama/Meta-Llama-3.1-8B)
Table 1 — CCS accuracy (mean $\pm$ sd over seeds)

dataset	answer (%)	n_answer	period (%)	n_period
comparisons	100 $\pm$ 00	30	100 $\pm$ 00	30
sp_en_trans	100 $\pm$ 00	30	100 $\pm$ 00	30
cities	100 $\pm$ 00	30	100 $\pm$ 00	30
amazon	95 $\pm$ 00	30	95 $\pm$ 00	30
imdb	88 $\pm$ 15	30	93 $\pm$ 01	30
ent_bank	98 $\pm$ 00	30	97 $\pm$ 09	30
snli	93 $\pm$ 00	30	90 $\pm$ 09	30
copa	72 $\pm$ 08	30	51 $\pm$ 02	30
rte	60 $\pm$ 02	30	57 $\pm$ 03	30

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Table 2 — loss-term ablations and alterations (answer token)

dataset	CCS	L_conf	L_cons	L_cons+a1	L_cons+a2	L_cons+a1+a2	CCS+a1+a2
comparisons	100 ± 00	100 ± 00	69 ± 13	69 ± 13	73 ± 15	64 ± 11	100 ± 00
sp_en_trans	100 ± 00	96 ± 07	66 ± 11	65 ± 11	72 ± 14	65 ± 11	100 ± 00
cities	100 ± 00	98 ± 06	67 ± 14	67 ± 13	70 ± 12	65 ± 10	100 ± 00
amazon	95 ± 00	95 ± 00	66 ± 10	66 ± 11	75 ± 11	66 ± 10	95 ± 00
imdb	88 ± 15	54 ± 01	58 ± 07	59 ± 06	65 ± 10	61 ± 08	90 ± 10

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Table 3 — tcPCA vs the spread of CCS solutions (period token)

dataset	CCS min	CCS med	CCS max	CRC-TPC	tcPCA	GCC
comparisons	100	100	100	100	100	100
sp_en_trans	100	100	100	100	100	100
cities	100	100	100	100	100	100
amazon	95	95	95	95	95	95
imdb	92	93	94	75	92	92
ent_bank	50	98	99	98	97	98
snli	55	93	95	91	93	93
copa	48	50	59	50	50	50
rte	48	56	65	56	55	56

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Table 5 — accuracy of individual principal components

dataset	token	PC1	PC2	PC3	PC4	PC5
comparisons	answer	100	58	55	43	54
sp_en_trans	answer	100	49	43	60	47
cities	answer	100	49	63	52	45
amazon	answer	95	52	46	56	52
imdb	answer	71	84	49	60	54
ent_bank	answer	97	57	51	50	50
snli	answer	92	50	60	62	56
copa	answer	72	64	50	56	54
rte	answer	60	54	63	48	62
comparisons	period	100	55	48	56	44
sp_en_trans	period	100	52	47	47	49
cities	period	100	53	52	49	52
amazon	period	95	48	45	54	46
imdb	period	75	79	57	53	60
ent_bank	period	98	54	50	47	60
snli	period	91	59	56	51	51
copa	period	50	64	48	60	51
rte	period	56	62	54	68	55

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Sign instability — how often the learned orientation came out backwards

dataset	answer flipped (%)	period flipped (%)
comparisons	53	52
sp_en_trans	55	51
cities	55	43
amazon	53	53
imdb	37	50
ent_bank	45	52
snli	50	52
copa	40	50
rte	42	57